



NCCER Module 22106



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Grades





Section One

Grade Planning

Objective

1. Explain how to plan and lay out a site for grading.
 - a. Explain the planning process for grading.
 - b. Explain how to mark and set grade stakes.



Performance Tasks

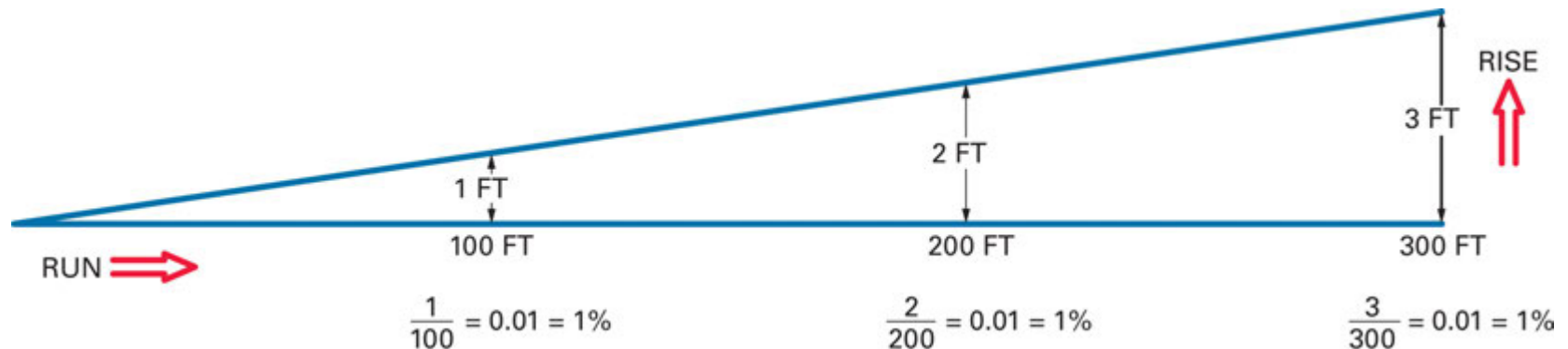
Under the supervision of an instructor, trainees should be able to do the following:

1. Identify types of stakes and markings on stakes.
2. Check horizontal and vertical distances of cut and fill slope stakes.



1.0.0 Grade Planning

- A grade is the rise or fall in the road, channel, or **natural ground**.
- For construction projects, engineers often express grade as a percentage above.
- Machine operators need to know the terms associated with grade, how they affect the construction project, and how to compute the rise and fall of grade.



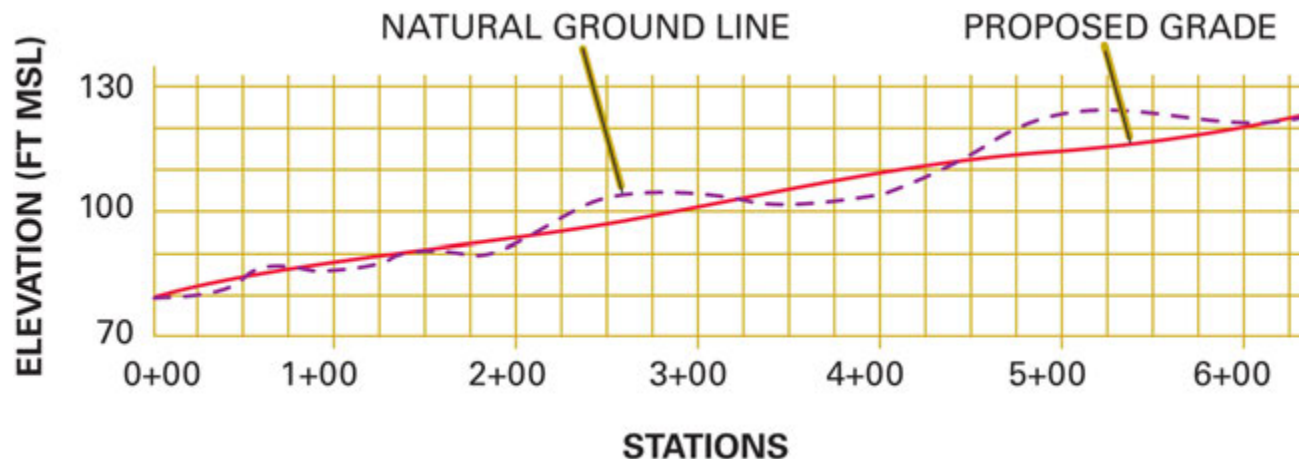
1.1.0 Site Planning and Layout

- The slope can contribute to **erosion**, which can damage the road or the foundation of a building.
- A survey crew marks the site according to the construction plans. The crew writes instructions on stakes and drives them into the ground to indicate whether earth needs to be removed from or added to an area to change the grade.
- Heavy equipment operators must learn to read the stakes, follow the directions written on them, and sometimes help set stakes.



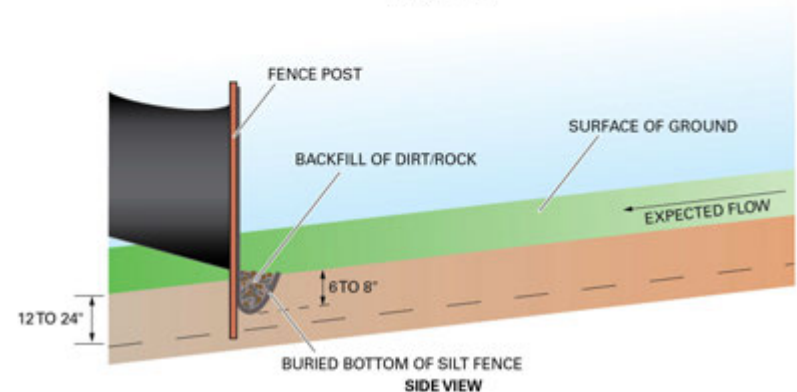
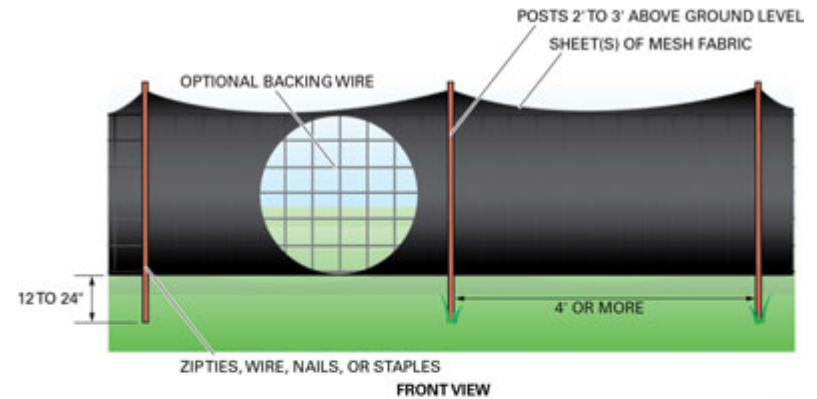
1.1.1 Profiles

- Architects or engineers create a profile on grid paper that shows the natural ground and proposed construction.
- **Station numbers** measure the distance from a reference point and are usually in increments of 100 feet.
- To build the roadway proposed in the profile below, operators need to cut earth from some areas and fill in other areas.



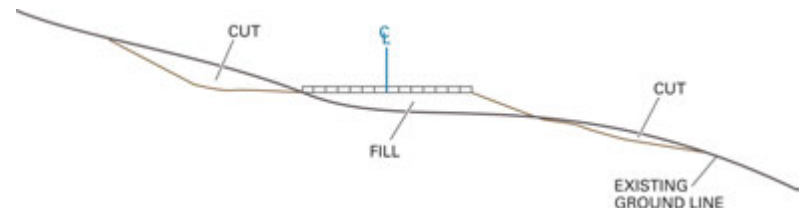
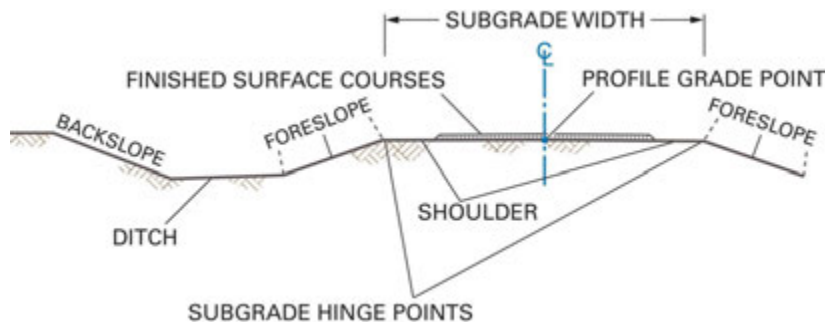
1.1.2 Erosion Controls

- State and federal guidelines require erosion to be put in place to keep sediment from flowing into nearby streams or properties.
- Silt fencing is the most common erosion control method.



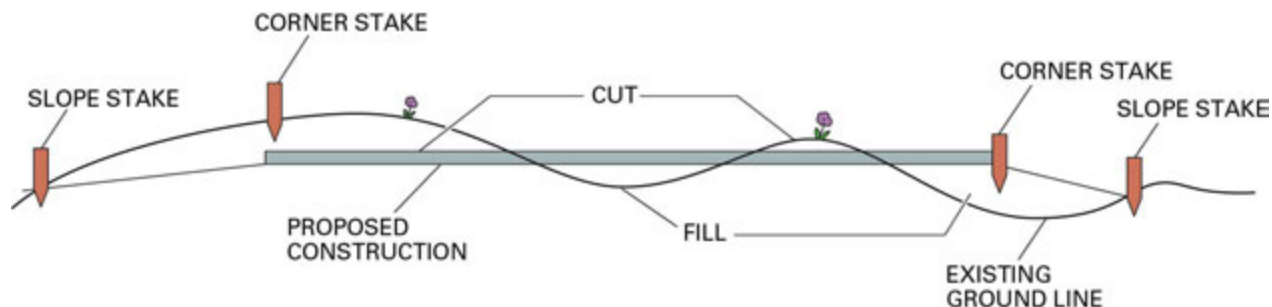
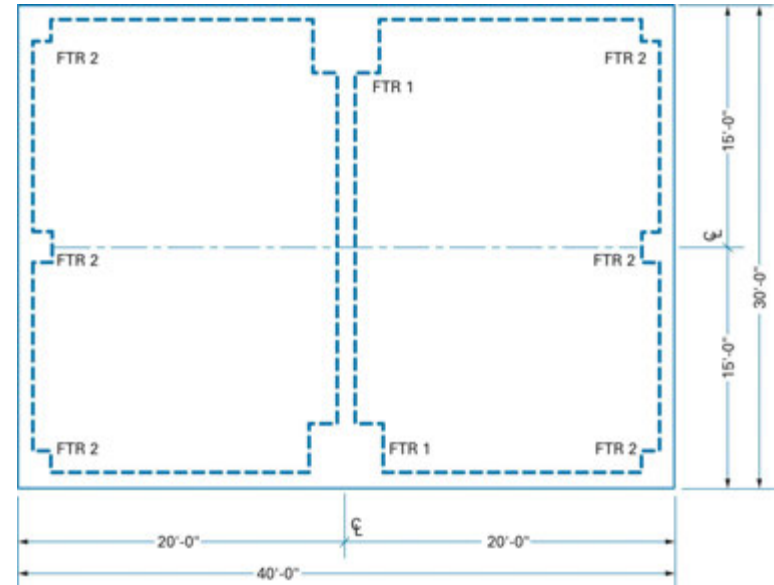
1.1.3 Roadways

- Workers apply the road foundation (called aggregate) and the road surface onto the **subgrade**.
- The edges of the subgrade where the downward slope increases are called shoulder points, **hinge points (HP)**, or the edge of road bed.
- Natural land may need to be cut and filled prior to road construction.



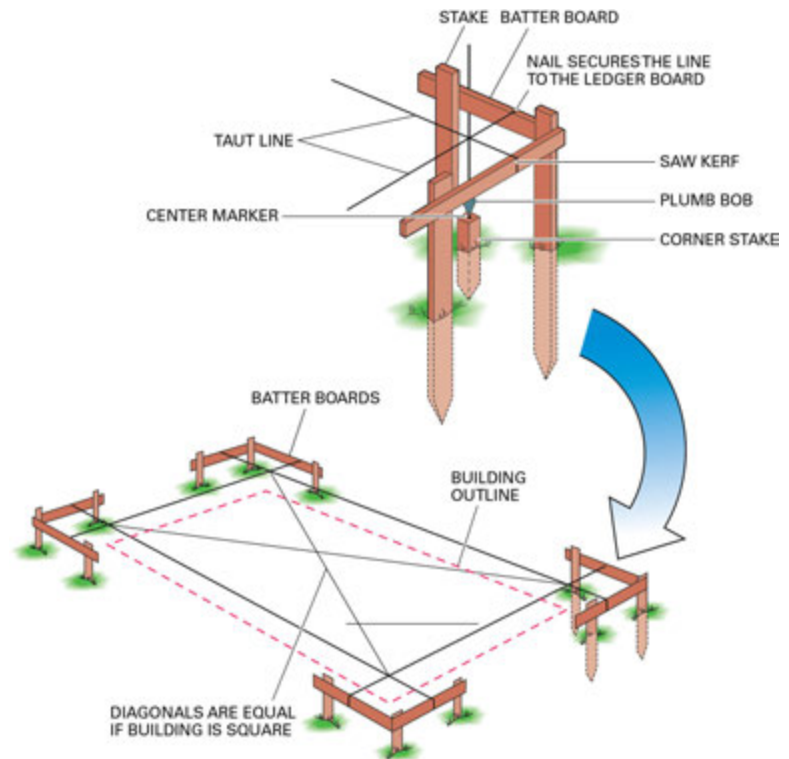
1.1.4 Buildings (1 of 2)

- A simple monolithic slab is a concrete slab formed in a single pour with no seams.
- To prepare a site for a monolithic slab, workers must level the area.



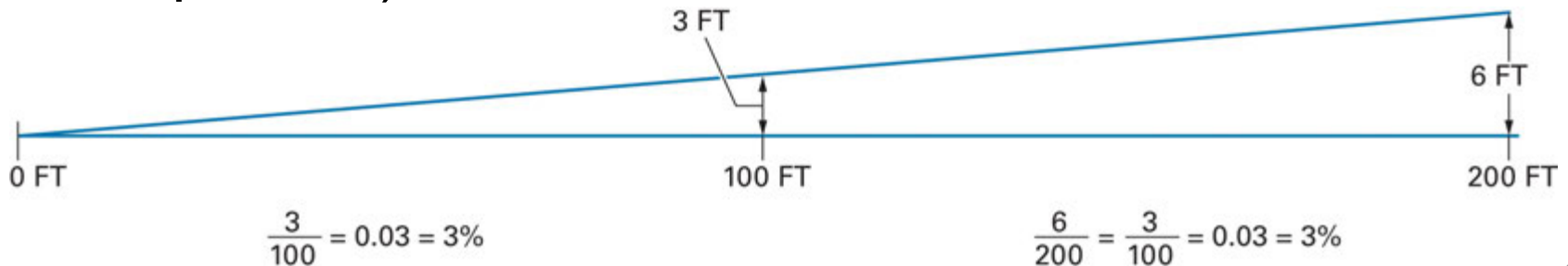
1.1.4 Buildings (2 of 2)

- To ensure that the building pad will be level and true, corner stakes must be set and batter boards must be used to establish building lines.
- Batter boards are used to set a reference **elevation**, such as the finish floor level of the building, as well as building lines.
- If you are working on a project and see stakes that you do not understand, ask for help before beginning the grading work.



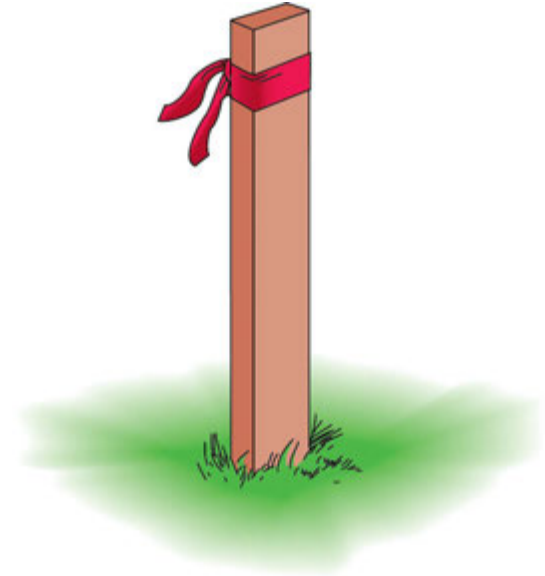
1.1.5 Grade Computation

- The slope of the grade is the **ratio** of vertical change in elevation (rise) for a given horizontal distance (run).
- When a grade goes up, it is a rising grade and includes a plus sign (+). A roadway that rises 3 feet over 100 horizontal feet of travel has a grade of +3.00% (plus three percent).
- When a grade goes down, it is a falling grade and includes a minus sign (–). A roadway that falls 3 feet over 100 horizontal feet of travel has a grade of –3.00% (minus three percent).

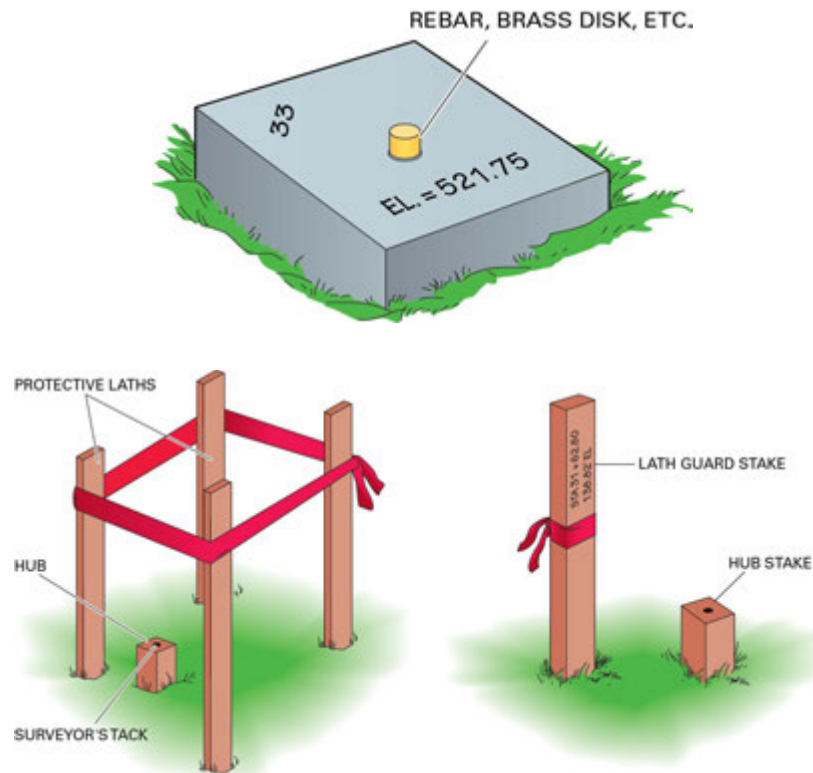


1.2.0 Marking and Setting Grade Stakes (1 of 3)

- Before grading work begins, the survey crew marks the area with stakes.
- Letters, numbers, and symbols on each stake indicate whether the area needs cutting or filling.
- Stakes that mark underground utilities use the standard underground color code.
- Grade stakes serve as reference points for cuts, fills, and boundaries.



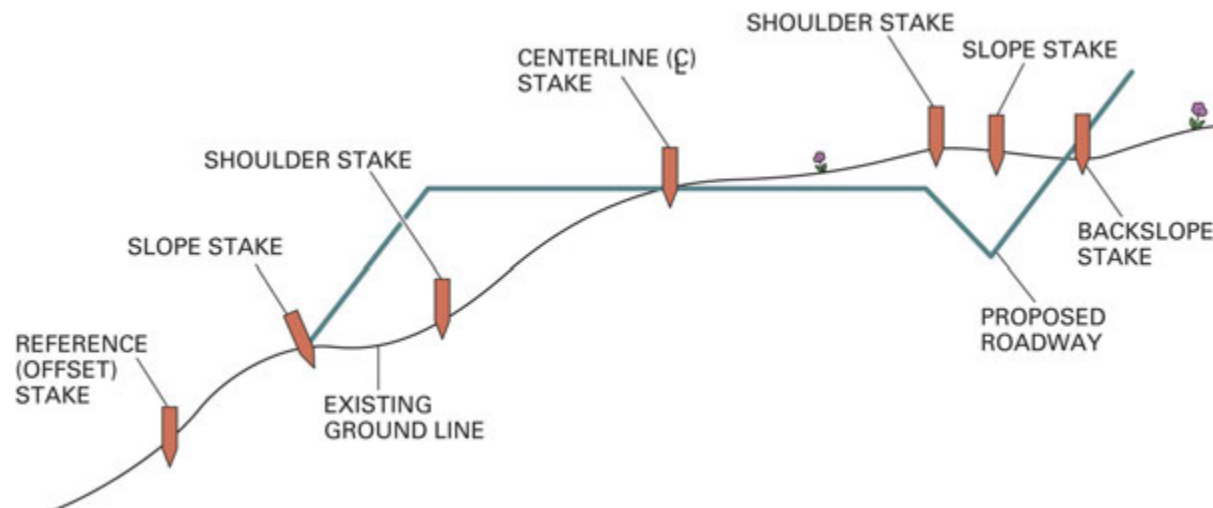
1.2.0 Marking and Setting Grade Stakes (2 of 3)



- A permanent benchmark can be a fixed object with a known position.
- Permanent benchmarks can be markers set in concrete and marked with an identifier and the elevation.
- **Temporary benchmarks (TBM)** are stakes, **hubs**, or markers placed by the survey crew in relation to the position of a permanent benchmark.

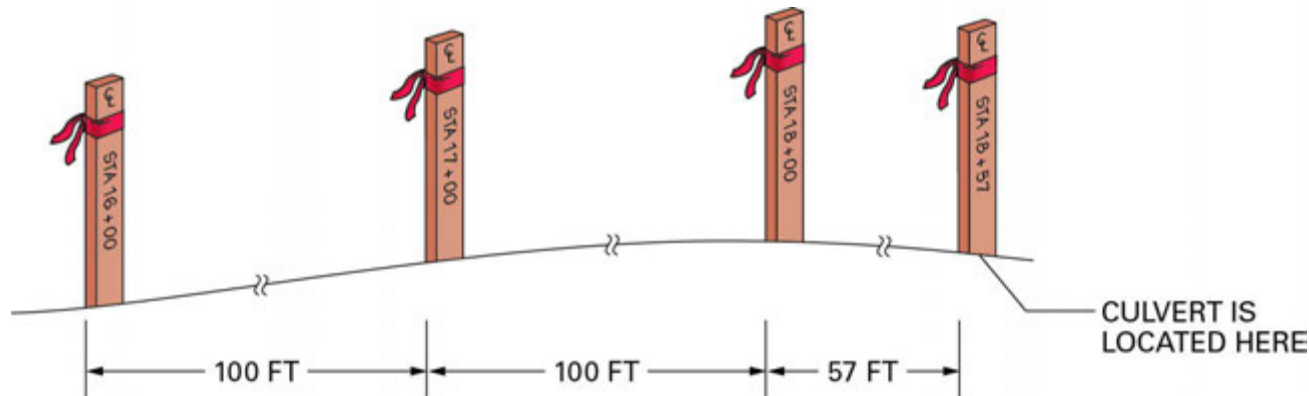
1.2.0 Marking and Setting Grade Stakes (3 of 3)

- Do not disturb or damage temporary benchmarks.
- For the heavy equipment operator, construction-site preparation has three phases: initial clearing, rough grading, and final grading.
- The survey crew sets rough boundary stakes for the clearing crew.
- The survey crew then marks the site for rough grading.



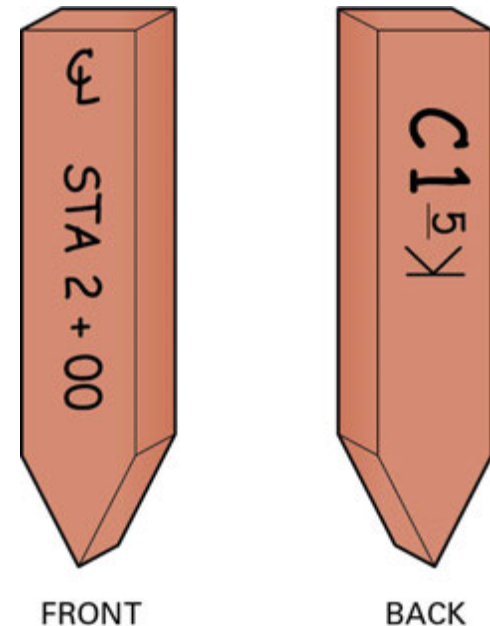
1.2.1 Center Line Stakes (1 of 2)

- Center line stakes mark the center of the proposed roadway and are marked with letters indicating cut and fill information.
- The station number shows the distance the stake is from a known location in 100-foot increments.
- When a culvert or the beginning of a curve is to be located between full stations, another center line stake is used.



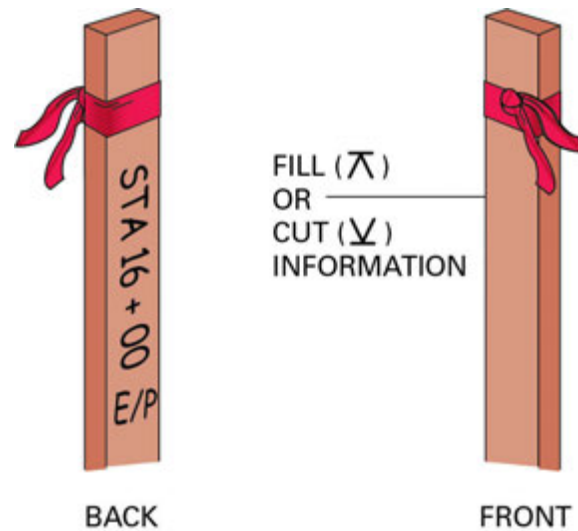
1.2.1 Center Line Stakes (2 of 2)

- The reverse side of the stake indicates cut or fill information. The letter C indicates a cut, and the letter F indicates fill.
- The **crow's foot** symbol points up for cut and down for fill.
- Measurements are written in the decimal system.
- Rough grade stakes display feet and tenths of a foot.
- **Finished grade (FG)** stakes are marked in hundredths of a foot.



1.2.2 Edge of Pavement Stakes

- Edge of road stakes or markers mark the edges of the paved surface.
- They are set on the shoulder line at right angles to the center line marker.
- The same station numbers written on the center line marker also appear on the back of these stakes so that they may be easily matched to a station.



1.2.3 Slope Stakes

- Slope stakes locate the edge of the cut or fill at the natural ground line some distance from the roadway hinge point.
- They indicate the run:rise ratio required to form the required slope.
- When the slope ratio is absent, the grade should go from the bottom of the ditch or edge of the road to the base of the stake.



BACK



FRONT

1.2.4 Offset Stakes

- To avoid resurveying parts of the project to set new stakes after work begins, offset stakes act as **reference stakes (RS)**.
- These stakes are located outside the actual grade-work area.
- A circle around the distance from **baseline** number and the letters O/S identifies an offset stake.



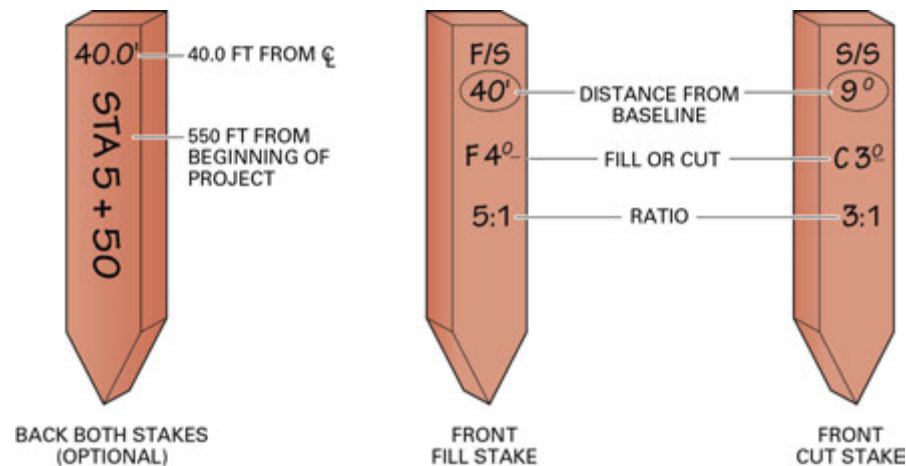
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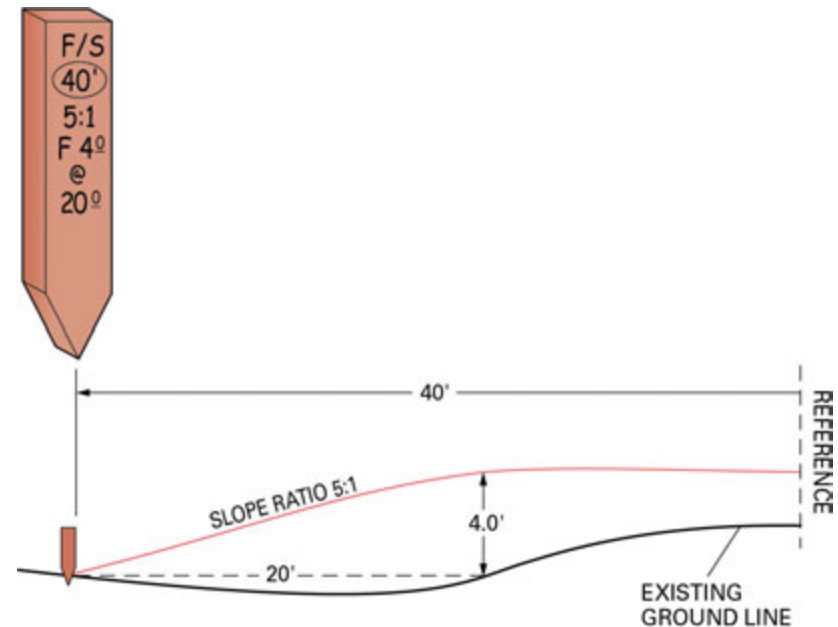
1.2.5 Interpreting Slope Stakes

- Slope stakes indicate the desired shape of a slope.
- A survey crew working for the engineering firm that designed the project will normally set out slope stakes.
- Slope stakes indicate distance from the reference (baseline, center line, or offset stake), cut or fill information, and slope ratio.



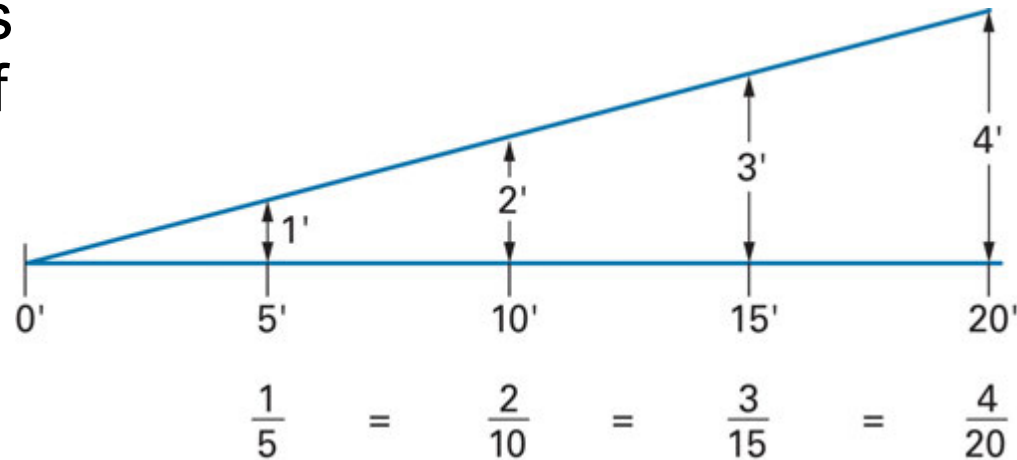
1.2.6 Fill Stakes (1 of 2)

- Fill stakes are placed where material must be added. They are marked with the letter F.
- The first number in the slope ratio on a cut or fill stake is the run.
- The second number is the rise/fall to grade for a given run.
- To figure the grade, it is the change of elevation (rise) for a given horizontal distance (run), expressed as a ratio or percentage.



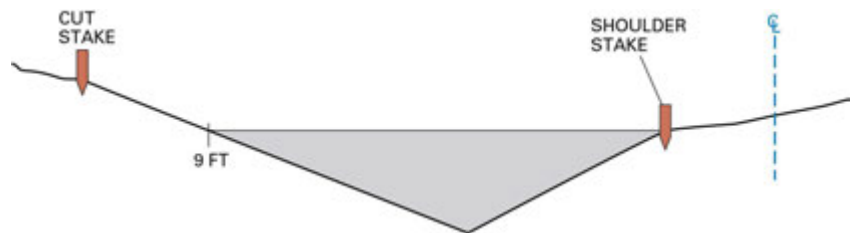
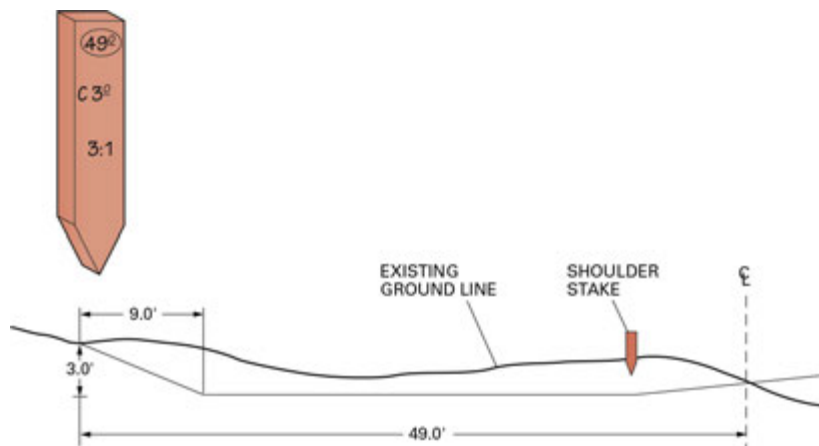
1.2.6 Fill Stakes (2 of 2)

The desired slope ratio is 5:1, so for every 5 feet of horizontal distance (run), 1 foot of fill (rise) is needed.



1.2.7 Cut Stakes

- This stake shows a cut of 3.0 feet (C 3⁰) and a slope ratio of 3:1.
- A ratio of 3:1 means that every 3 feet from the stake, the ground must drop 1 foot. To get to a 3.0-foot cut, the run must be 9.0 feet.
- After the initial 3.0-foot cut, the area will probably need excavating to form a ditch.



Wrap Up – Trade Terms (1 of 4)

Baseline: A real or imaginary line that is the reference for other measurements.

Cross section: A cutaway view of an object showing its shape and features perpendicular to its longitudinal axis or center line.

Crow's foot: A symbol on a stake or lath to indicate the vertical direction of the final grade from the existing grade at the location of the stake.

Elevation: The vertical distance of grade above or below mean sea level (MSL) or other reference point.

Erosion: The process of removing soil or rock by the action of rain, wind, flowing water, or other causes.



Wrap Up – Trade Terms (2 of 4)

Finished grade (FG): Any surface that has been cut or filled to the final elevation specified for that point in construction plans.

Grade work: Plotting and correction of irregularities of the ground needed to arrive at the desired grade and alignment.

Hinge points (HP): On cross-section diagrams, the points indicating where the road or shoulder subgrade ends and the slope steepens to an adjacent ditch or other area. Sometimes given the term catch point.

Hub: A point-of-origin stake that identifies a point on the ground. The top of the hub establishes the point from which soil elevations and distances are computed or a point to be trimmed to.



Wrap Up – Trade Terms (3 of 4)

Natural ground: The original ground elevation before any excavation has been done. Also called existing grade.

Ratio: Proportion; relationship in quantity, amount, or size between two or more things. For example, if a grade has a vertical rise of 1' for every 3' of horizontal run, the ratio of rise to run is 1:3.

Reference stakes (RS): The stakes from which measurements and grades are established.

Station numbers: Numbers on a roadway profile and surveying stakes indicating the longitudinal (lengthwise) distance of a point of a roadway from the starting or reference point. Station numbers usually represent 100' increments.



Wrap Up – Trade Terms (4 of 4)

Subgrade: The uppermost level of material placed in an embankment or left at cuts in the normal grading of a road bed. This becomes the foundation for aggregate and asphalt material.

Temporary benchmarks (TBM): A temporary point of known or assumed elevation from which surveyors can establish all their grades for a particular job.

Topographical: Having to do with the description and mapping of the natural landforms and man-made features of a geographical area.



1.0.0 Section Review Question 1

1. A side view of a planned highway is called its _____.
- a. cross section
 - b. profile
 - c. offset
 - d. natural ground



1.0.0 Section Review Question 1 Answer

1. A side view of a planned highway is called its _____.
- a. cross section
 - b. profile**
 - c. offset
 - d. natural ground



1.0.0 Section Review Question 2

2. Which instruction on a stake correctly represents the amount of fill at a location?
- a. 8+23
 - b. F (8)
 - c. F 4- $\frac{1}{2}$
 - d. F 4⁵



1.0.0 Section Review Question 2 Answer

2. Which instruction on a stake correctly represents the amount of fill at a location?
- a. $8+23$
 - b. F ⑧
 - c. F $4-\frac{1}{2}$
 - d. F 4^5



Next...

Section 2.0.0

Performing Site Measurements



*Read Sections 2.0.0 through 2.3.2 and
complete the Section Review.*